INCH - POUND

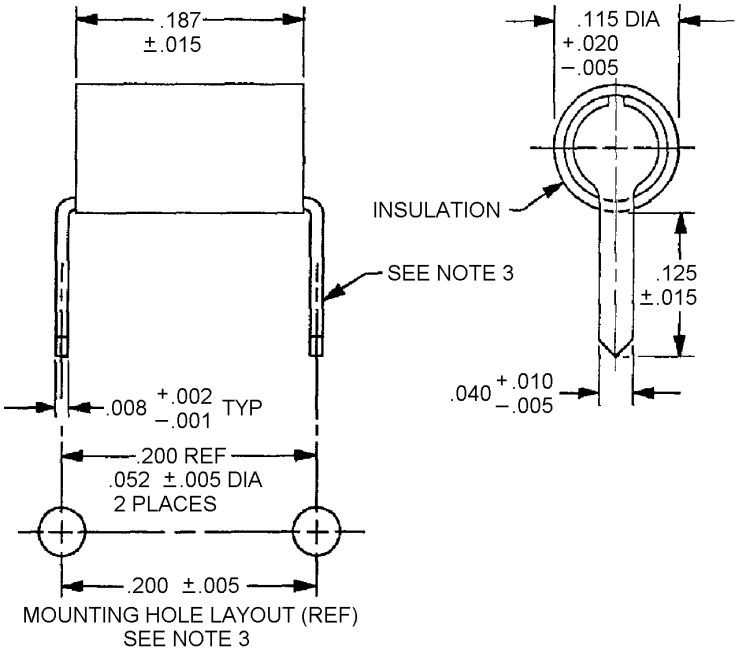
MIL-DTL-39024/18F
w/AMENDMENT 1
3 June 2020
SUPERSEDING
MIL-DTL-39024/18F
22 September 2008

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, TEST POINT TYPE, PRINTED WIRING TYPE; MICRO-MINIATURE, SINGLE TEST POINT (2-LEG MOUNTING), LOW VOLTAGE, .080 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-39024.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .040 | 1.02 |
| .002 | 0.05 | .052 | 1.32 |
| .005 | 0.13 | .115 | 2.92 |
| .008 | 0.20 | .125 | 3.18 |
| .010 | 0.25 | .187 | 4.75 |
| .015 | 0.38 | .200 | 5.08 |
| .020 | 0.51 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Mounting to fit .052 (1.32 mm) dia holes (2 holes), and lead spacing shall be such to meet .200 ± .005 mounting hole layout requirements.
4. Dimensions are valid after plating only.

FIGURE 1. Dimensions and configuration.



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TABLE I. Dash number and identification data.

| Dash number | Insulation color in accordance with MIL-STD-104 |
|---|---|
| 01 | White |
| 02 | Yellow |
| 03 | Brown |
| 04 | Red |
| 05 | Orange |
| 06 | Black |
| 07 | Green |
| 08 | Blue |
| 09 | Gray |
| 10 | Purple |

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Contact: Contact shall be phosphor bronze in accordance with ASTM B139/B139M.

Insulation: Material shall be polytetrafluoroethylene in accordance with MIL-I-22129 or equal.

Test probe: 0.080 ± 0.001 inch in diameter (connector shall accept test probe from either end).

Operating voltage: 1,000 volts rms, maximum.

Operating temperature range: -65°C to +200°C.

Insulation resistance: Test circuit in MIL-DTL-39024 shall be used.

Dielectric withstanding voltage: Test circuit in MIL-DTL-39024 shall be used.

Test voltage (sea level): 1,000 volts rms, 60 hertz, shall be applied for 15 seconds.

Test voltage (50,000 feet): 350 volts rms, 60 hertz, shall be applied for 15 seconds.

Insertion and withdrawal forces:

Insertion force: 10.0 pounds (max).

Withdrawal force: 0.5 pound (min).

Resistance to test probe damage: ½ inch-pound moment shall be used for insertion depth of 3/16 inch and ¼ inch.

Shock (specified pulse): Method 213, test condition I of MIL-STD-202.

Vibration, high frequency: 15 g's at 10 to 2,000 Hz, method 204, test condition B of MIL-STD-202.

Durability: Not applicable.



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Identification marking:

Part or Identifying Number (PIN): M39024/18-(dash number from table I).

Operating conditions:

These connectors are designed for 1/16 inch nominal thickness printed wiring board (PIN M39024/18-01 thru 10)

Supersession data: This specification includes the requirements of contractor (90348) specification 2134873; contractor (05869) specification 716007.

Qualification: Not applicable.

First article inspection. First article inspection shall consist of Group A and B inspections. MIL-DTL-39024 paragraph 4.5 through 4.5.3.1.3 and Tables III through VII as applicable.

Conformance inspection. Inspection of successive lots for delivery shall consist of Group A inspections. MIL-DTL-39024 paragraph 4.5 through 4.5.3.1.3 and Tables III through VII as applicable.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-39024, this document references the following:

ASTM B139/B139M
MIL-I-22129
MIL-STD-104
MIL-STD-202-204
MIL-STD-202-213

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2020-084)

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC
Air Force – 19

NOTE: The activities listed above were interested in this document as of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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